

## LINDA, Antarctica

WMO: 897690

Lat: 78.394S

Lon: 168.446E

Elev: 135

StdP: 14.62

Time Zone: 12.00 (NZL)

Period: 96-12

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |       | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|-------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|-------|-----------------------|------|-----|
|               |            |       | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |       |                       |      |     |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB  | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)   | (n)                   | (o)  | (p) |
| 8             | -59.9      | -55.7 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 58.4                  | -9.1 | 51.7 | -12.4 | 4.9                   | 190  | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 12            | 13.6                   | 34.5            | N/A  | 31.3 | N/A  | 28.6 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 5.2                  | 300  |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |  |  |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|--|--|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |  |  |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |  |  |
| (4) 47.6          | 42.0 | 36.8 | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
| (5)               |      |      | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | -9.3   | 20.4  | 5.9   | -12.2 | -20.1 | -23.7 | -22.7 | -25.9 | -28.2 | -23.5 | -11.8 | 9.2   | 22.2  |  |  |
|                                                                        | DBStd       | 21.26  | 5.56  | 8.40  | 9.89  | 14.03 | 14.89 | 14.36 | 14.66 | 13.89 | 13.62 | 9.23  | 8.47  | 5.23  |  |  |
|                                                                        | HDD50       | 21632  | 918   | 1234  | 1927  | 2102  | 2285  | 2182  | 2353  | 2424  | 2206  | 1915  | 1223  | 863   |  |  |
|                                                                        | HDD65       | 27107  | 1383  | 1654  | 2392  | 2552  | 2750  | 2632  | 2818  | 2889  | 2656  | 2380  | 1673  | 1328  |  |  |
|                                                                        | CDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 15.0   | 11.0  | 11.8  | 12.8  | 16.0  | 16.0  | 17.2  | 16.9  | 16.3  | 17.9  | 17.1  | 15.2  | 11.1  |  |  |
|                                                                        | PrecAvg     | 6.1    | 0.6   | 0.6   | 0.6   | 0.5   | 0.5   | 0.5   | 0.4   | 0.6   | 0.5   | 0.4   | 0.4   | 0.5   |  |  |
| (15) Precipitation                                                     | PrecMax     | 9.5    | 1.6   | 1.3   | 1.3   | 1.4   | 1.1   | 1.5   | 1.4   | 1.9   | 1.3   | 1.0   | 1.2   | 1.1   |  |  |
|                                                                        | PrecMin     | 2.0    | 0.1   | 0.2   | 0.1   | 0.2   | 0.0   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.0   |  |  |
|                                                                        | PrecStd     | 1.6    | 0.4   | 0.3   | 0.2   | 0.3   | 0.3   | 0.4   | 0.3   | 0.4   | 0.3   | 0.3   | 0.3   | 0.3   |  |  |
|                                                                        | PrecStd     | 1.6    | 0.4   | 0.3   | 0.2   | 0.3   | 0.3   | 0.4   | 0.3   | 0.4   | 0.3   | 0.3   | 0.3   | 0.3   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 38.9  | 27.6  | 14.5  | 18.6  | 10.6  | 11.3  | 8.8   | 10.2  | 10.1  | 12.8  | 27.3  | 41.1  |  |  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 2%          | DB     | 34.0  | 23.8  | 10.2  | 8.1   | 4.6   | 8.1   | 4.9   | 4.7   | 5.4   | 8.7   | 24.9  | 35.7  |  |  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 5%          | DB     | 30.6  | 20.7  | 6.9   | 3.4   | 1.0   | 3.9   | 0.4   | -0.6  | 1.1   | 5.0   | 22.5  | 32.6  |  |  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 10%         | DB     | 28.2  | 17.8  | 3.1   | -0.9  | -2.7  | -1.6  | -4.8  | -6.7  | -3.5  | 1.6   | 20.3  | 30.1  |  |  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 13.6  | 13.6  | 14.3  | 15.4  | 16.3  | 16.8  | 16.1  | 17.6  | 16.7  | 16.2  | 12.9  | 13.6  |  |  |
|                                                                        |             | MCDBR  | 18.7  | 16.2  | 13.6  | 16.5  | 17.2  | 16.7  | 16.6  | 16.9  | 17.0  | 18.8  | 14.4  | 19.2  |  |  |
|                                                                        |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.235  | 0.214 | 0.169 | 0.113 | N/A   | N/A   | N/A   | N/A   | N/A   | 0.161 | 0.193 | 0.215 | 0.234 |  |  |
|                                                                        | taud        | 2.175  | 2.189 | 2.125 | 1.878 | N/A   | N/A   | N/A   | N/A   | N/A   | 2.084 | 2.188 | 2.200 | 2.180 |  |  |
|                                                                        | Ebn at Noon | 303    | 282   | 239   | 21    | 0     | 0     | 0     | 0     | 0     | 237   | 293   | 312   | 312   |  |  |
|                                                                        | Edh at Noon | 33     | 25    | 15    | 1     | 0     | 0     | 0     | 0     | 0     | 15    | 25    | 32    | 35    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 2555   | 1516  | 486   | 33    | 0     | 0     | 0     | 0     | 0     | 257   | 1136  | 2399  | 3006  |  |  |
|                                                                        | RadStd      | 56     | 54    | 22    | 2     | 0     | 0     | 0     | 0     | 0     | 14    | 30    | 70    | 76    |  |  |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (5 neighbors) | +1.44 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page